Project Assignment

Model Predictive Control for Racing Line Tracking Using a Dynamic Bicycle Model in F1TENTH Gym

Within the scope of this project, a Model Predictive Control (MPC) algorithm shall be developed to enable precise tracking of a pre-computed optimal racing line. The system will be implemented in ROS 2 and tested in the F1TENTH Gym simulation environment. The foundation is a dynamic bicycle model that accurately represents the F1tenth vehicle's dynamics including lateral tire forces and slip angles.

The objective is to implement a real-time capable MPC controller that minimizes tracking errors while respecting physical constraints such as maximum steering angle, velocity limits, and tire friction limits. The focus is on the mathematical formulation of the MPC optimization problem, implementation using appropriate optimization libraries, and integration with the F1TENTH Gym environment for testing and validation.

Tasks

1. Familiarization with Model Predictive Control theory, dynamic bicycle models, and the F1TENTH Gym simulation environment.
2. Formulation of the MPC optimization problem for racing line tracking, including objective function (tracking error minimization) and constraints (vehicle dynamics, actuator limits, friction circle).
3. Implementation of the MPC controller in ROS 2 using an optimization library (e.g., CasADi, ACADO, or similar) with real-time capability considerations.
4. Integration with the F1TENTH Gym environment for closed-loop simulation and testing of the controller.
5. Tuning of MPC parameters (prediction horizon, control horizon, weighting matrices) and validation of tracking performance at different velocities.
6. Documentation of the MPC formulation, implementation details, and comparative evaluation of tracking performance (e.g., tracking error, computational time, handling of constraints).

The final project has to be presented as a 20-minute live code demonstration with additional 10 minutes for questions.